The Kettle vs. The Supercharger: Decoding Electrical Power

GCSE Physics · High-Power Engineering & Energy Transfer



2,300 W

A standard kitchen kettle draws 2,300 Watts. An ultra-rapid EV supercharger delivers 350,000 Watts.

How many boiling kettles of electrical power are flowing simultaneously through that single charging cable?

~152 simultaneous kettles.



350,000 W

Learning Goal: Connect Watts, Volts, Amps, and Joules to solve real-world thermal limits in high-power vehicle engineering.

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System Diagnostics: Knowledge Retrieval

What formula links electrical power, current, and potential difference?

[Success Cue: P = V x I]

P = V x I

What is the unit of electrical power, and what does 1 Watt equal in energy per second?

[Success Cue: Joules per second]

Watt (W) = 1 Joule per second (1 J/s)

A domestic kettle operates on 230 V mains and draws 10 A. Calculate its power rating.

[Success Cue: P = 230 x 10]

2,300 W (2.3 kW)

What formula links total energy transferred, power rating, and operating time?

[Success Cue: Energy = Power x time]

E = P x t

How much electrical energy is transferred when a 2.3 kW kettle boils water for 200 seconds?

[Success Cue: E = 2300 x 200]

460,000 J (460 kJ)

If electrical current through a charging cable doubles, by what factor does resistive heat loss increase?

[Success Cue: P = I squared x R relationship]

Power quadruples (increases 4x)

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Power as the Rate of Energy Transfer (P = E / t)

Key Takeaway: Power limits your charging speed, not the battery's total capacity.

$$P = E / t$$

(Power in Watts = Energy in Joules divided by Time in seconds)

Domestic 3-Pin Plug	Dedicated Home Wallbox	Motorway Rapid DC Charger
Power **2.3 kW (1 Kettle)**	Power **7.4 kW (3 Kettles)**	Power **150 kW (65 Kettles)**
Time **~32.6 hours**	Time **~10.1 hours**	Time **~30 minutes**
Context Overnight + Full Day	Context Single Overnight Charge	Context Quick Coffee Stop
5%	50%	100%

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Inside the Cable: Current & Voltage (P = I x V)

How do engineers deliver 150 kW without melting the charging cable?

⚠ Option A: Crank the Current (I)

Thick, heavy, unmanageable copper cables. Extreme resistive heat.

Delivering 150,000 W on standard 230 V.

I = P / V

150,000 / 230 = **652 A**

(Cable overheats and melts!)



P = I x V

Option B: Boost the Voltage (V)

Thin, manageable, lightweight cables. Why modern high-performance EVs use an 800 V architecture.

Delivering 150,000 W on 800 V.

I = P / V

150,000 / 800 = **187.5 A**

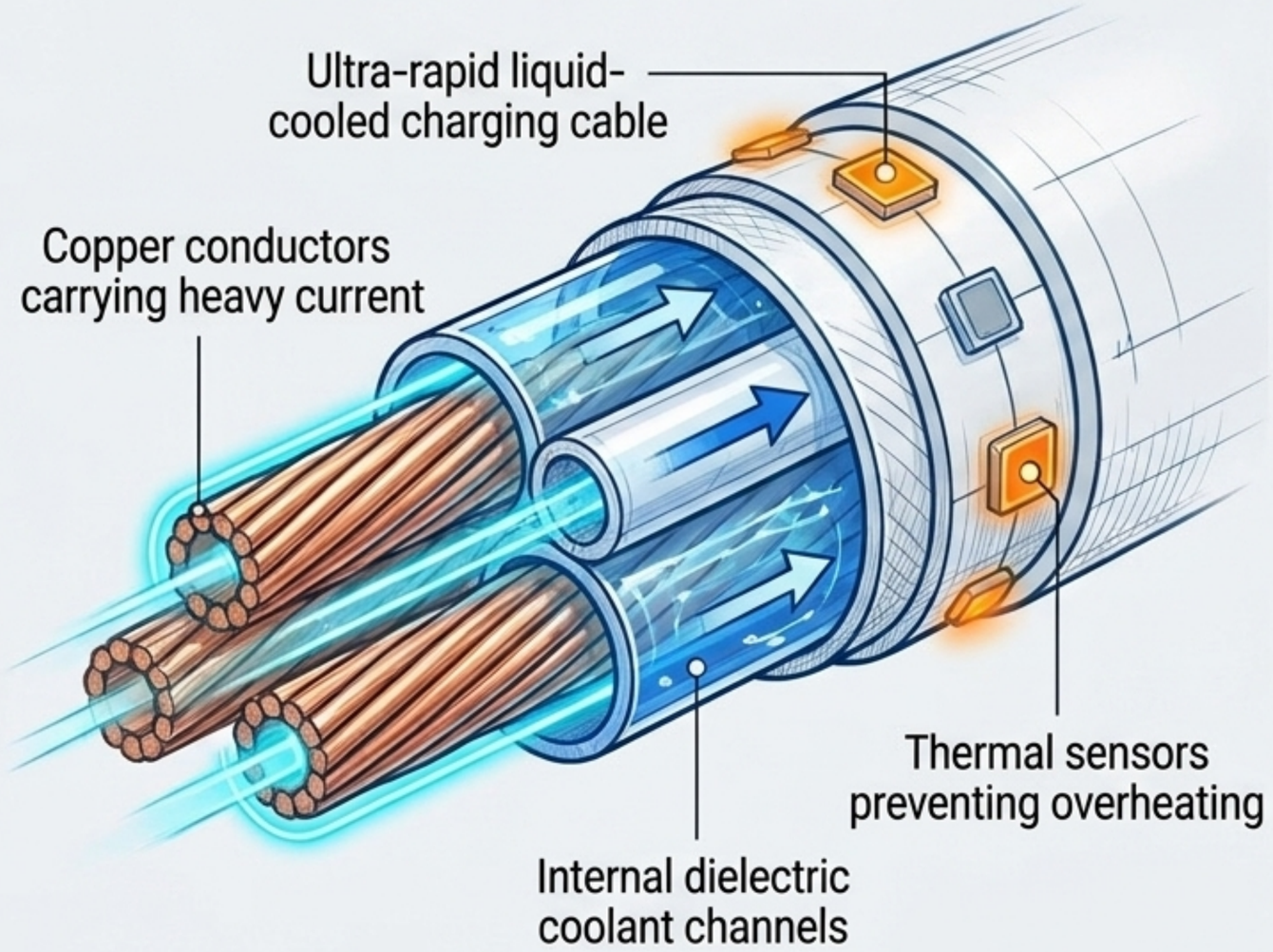
(Safe and manageable).



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The Thermal Limit: The Heat Problem (P = I^2 x R)

Core Concept: Doubling current quadruples heat loss. High voltage is the only solution.



$$P = I^2 \times R$$

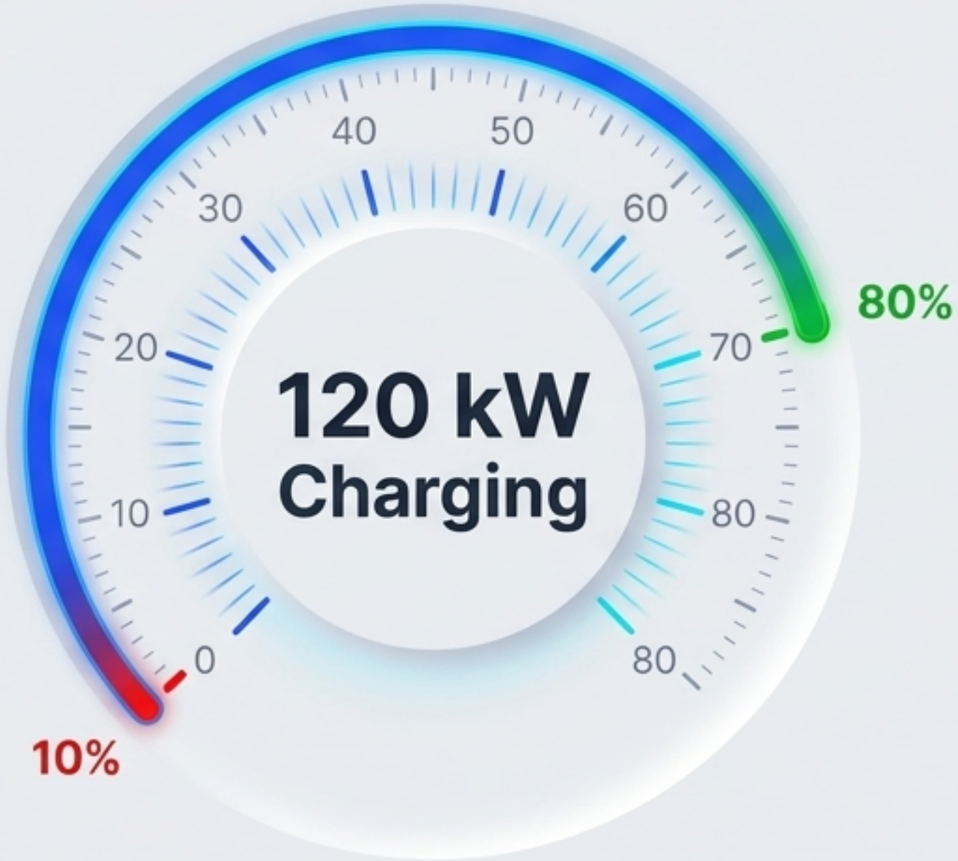
Because current (I) is squared, increasing flow is highly punishing.

Double the current (2x)	→	Heat loss **increases 4x** (2^2)
Triple the current (3x)	→	Heat loss **increases 9x** (3^2)

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Telemetry Check: Student Calculation Challenge

Scenario: A driver arrives at a motorway service station with 10% charge on a 60 kWh battery and needs to reach 80% to complete the journey safely. The charger delivers a steady 120 kW.



Step 1: Calculate the Energy required.

Target is 80% - 10% = 70% of battery capacity.

0.70 x 60 kWh

42 kWh

Step 2: Calculate the Time required.

$$t = E / P$$

42 kWh / 120 kW = 0.35 hours.

0.35 x 60 minutes.

21 minutes

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Unpacking the Trap: Energy vs. Power



POWER (The Flow Rate)

Units: Watts (W) or Kilowatts (kW)

Definition: The rate at which energy is transferred (P = E/t).

How wide the hose is.



ENERGY (The Total Volume)

Units: Joules (J) or Kilowatt-hours (kWh)

Definition: The total amount of work done or stored.

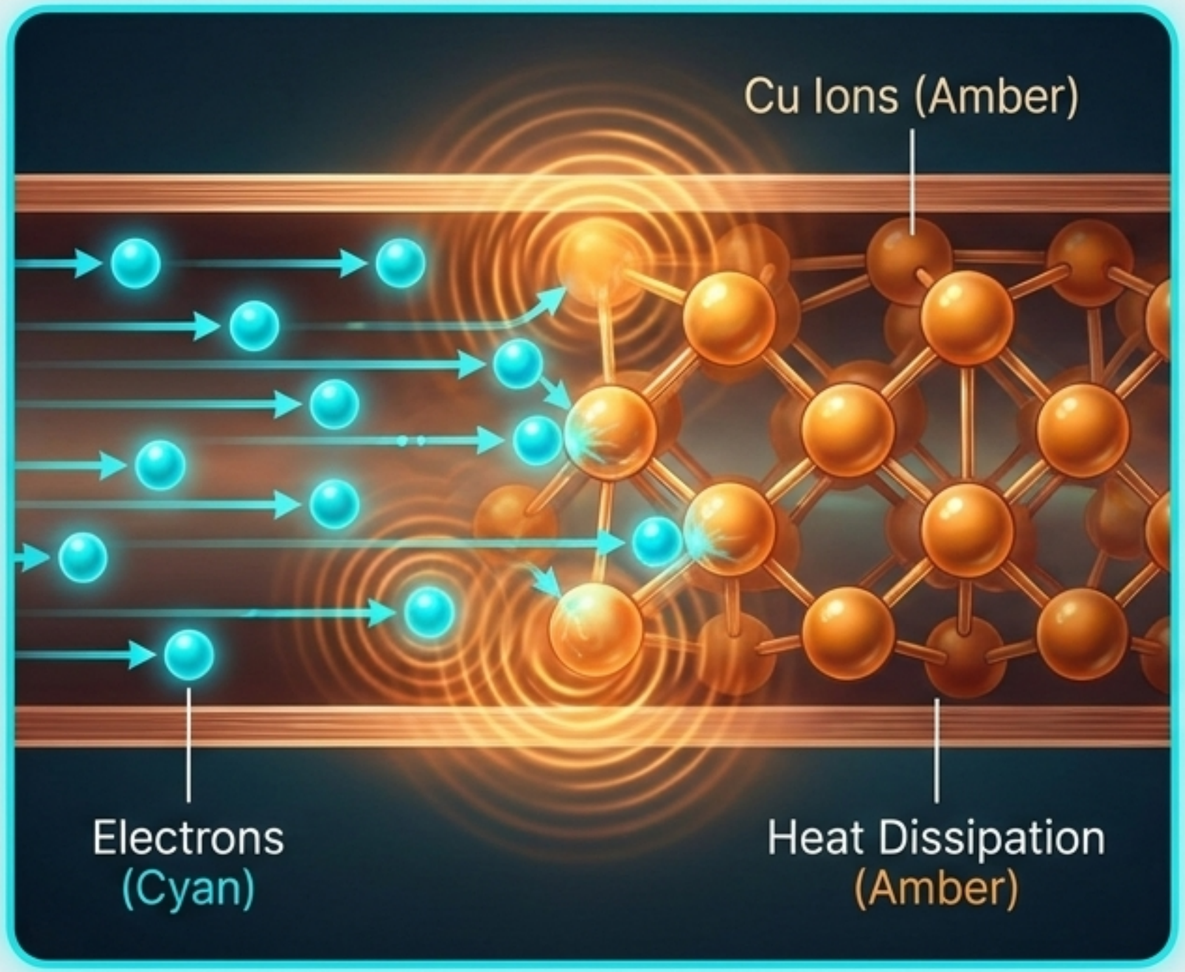
How much water is in the bucket.

You cannot put ‘kW’ into a battery. You put ‘kWh’ (or Joules) into a battery by applying a ‘kW’ flow rate over time.

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The Physical Barrier: Resistance (V = I x R)

Core Concept: Why does a cable heat up in the first place? Ohm's Law in action.



The Physics

Resistance is the collision between flowing electrons and the metal ions.

More collisions = more wasted thermal energy (heat).

The Golden Triangle



Engineering Fixes

1. **Increase cross-sectional area** (thicker wire)
   → Decreases R (more paths for electrons).
2. **Decrease length** (shorter cable)
   → Decreases R (fewer collisions).

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Efficiency: Where Does the Wasted Energy Go?

Efficiency = (Useful Energy Output / Total Energy Input)



Key Insight: No system is 100% efficient. Wasted energy is not destroyed; it is simply stored in ‘less useful ways’ (dissipated to the surroundings).

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System Overview: The Power Scale Matrix

Core Concept: Synthesizing voltage, current, and time across three engineering scales.

	The Kitchen Kettle	Home EV Wallbox	Ultra-Rapid Supercharger
Operating Voltage (V)	230 V	230 V	800 V
Max Current (I)	10 A	32 A	**437 A**
Power Output (P)	2.3 kW	7.4 kW	350 kW
Time to transfer 10 kWh (t = E/P)	4.3 hours	1.3 hours	1.7 mins

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Scaling Up: The National Grid

If 650 A melts a single EV cable, what happens when you try to send power for an entire city across the country?



The National Grid is an efficient way to transfer energy because it steps up the voltage to massive levels (e.g., 400,000 **V**) specifically to keep the current incredibly low, minimizing $\mathbf{I}^2 \times \mathbf{R}$ heat loss in the transmission lines.

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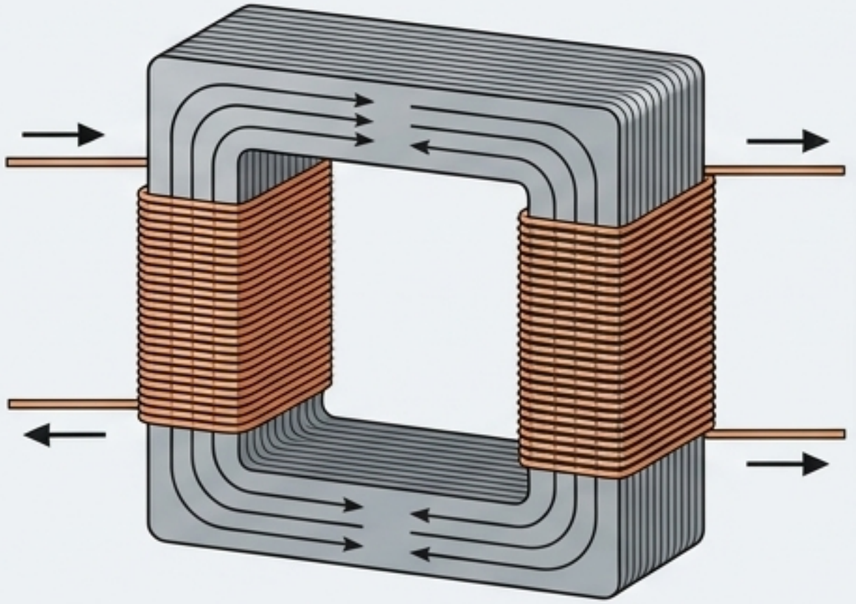
Transformers: The Ultimate P = I x V Application

Core Concept: Manipulating **V** and **I** to cheat the heat.

Input (Primary)

25,000 V x **4,000 A** =

100 MW Power



Output (Secondary)

400,000 V x **250 A** =

100 MW Power

Notice the Conservation of Energy: Power (P) stays exactly the same (100 MW).
Voltage (V) goes up by **16x**, so Current (I) must drop by **16x.**

Because heat loss is $I^2 \times R$, dropping the current by 16x reduces heat loss by 256 times (16^2)!

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System Warnings: The 3 Exam Pitfalls

Core Concept: Navigating the most common unit and mathematical traps in OCR physics.

Warning 1: The Time Trap



Power is Joules per second. If an exam gives you charging time in hours, you MUST multiply by 3,600 before using **P** = **E**/**t**.

Warning 2: The Prefix Trap



Standard Form prefixes: **k** = 1,000 (10^3), **M** = 1,000,000 (10^6). A 350 k**W** charger must be written as 350,000 **W** before being placed into the **P** = **I** x **V** equation.

Warning 3: Significant Figures

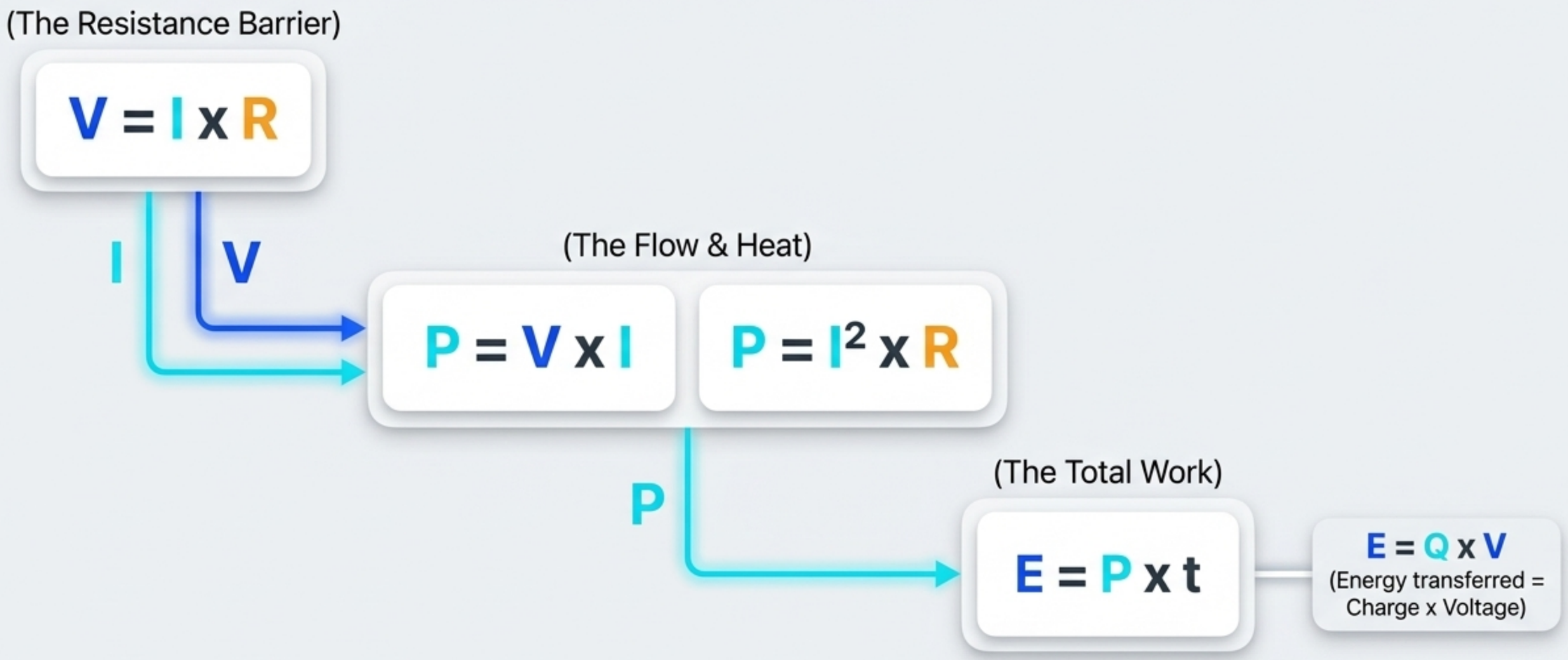


Always round your final answer to the lowest number of significant figures given in the original question prompt.

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The Synthesis Map: Interlocking Equations

Takeaway: Don't memorize isolated facts. Understand how the variables flow into one another.



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Plenary Knowledge Check

Core Concept: Consolidation of the macro-to-micro journey.

Why do ultra-rapid chargers use 800V instead of standard 230V mains?

High Voltage allows lower Current for the same Power, drastically reducing $I^2 \times R$ heat loss in the cable.

What system does the National Grid use to achieve this exact same effect over long distances?

Step-up Transformers.

A charger operates at 50,000 W for 10 seconds. How many Joules of energy are transferred?

500,000 J (or 500 kJ).

Next time you boil a kettle, remember: there's an engineer figuring out how to control 152 of them in the seernet at and engineer figuring out the control 152 of them in the palm of your han

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